THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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ANSWER KEY & MARKING SCHEME SET A

Chapter 5: **Circles** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 24 cm	half-chord $= \sqrt{13^2 - 5^2} = 12$, so chord $= 2 \times 12 = 24$
2	**(d)** 48 cm	half-chord $= \sqrt{25^2 - 7^2} = 24$, so chord $= 2 \times 24 = 48$
3	**(a)** 10 cm	$r = \sqrt{8^2 + 6^2} = 10$
4	**(c)** 15 cm	$r = \sqrt{9^2 + 12^2} = 15$
5	**(a)** 8 cm	$d = \sqrt{10^2 - 6^2} = 8$
6	**(c)** 65°	angle at the centre is double the angle on the circle
7	**(c)** 70°	opposite angles of a cyclic quadrilateral add to 180°
8	**(d)** 56°	central angle is double
9	**(a)** equal	equal chords $\Leftrightarrow$ equal distances
10	**(a)** concyclic points	definition

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Half-chord $= \sqrt{17^2 - 8^2} = \sqrt{225} = 15$ cm; chord $= 30$ cm.	1+1
12	Half-chord 20; $r = \sqrt{20^2 + 21^2} = \sqrt{841} = 29$ cm.	1+1
13	$\angle A + \angle C = 180^\circ$ and $\angle B + \angle D = 180^\circ$, so yes — opposite angles add up to 180° and the quadrilateral is cyclic.	1+1
14	Half of the central angle: 35°.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let $OM \perp AB$. In right triangles OMA and OMB: $OA = OB$ (radii), OM common, $\angle OMA = \angle OMB = 90^\circ$ ⇒ congruent (RHS) (2). So $AM = MB$ (1).	2+1
16	The arc not containing D subtends a straight angle 180° at the centre (1). The angle at D is half of this (1): $\angle ADB = \frac{1}{2} \times 180^\circ = 90^\circ$ (1).	1+1+1
17	Arcs BCD and BAD together make the whole circle. $\angle BAD = \frac{1}{2}$(reflex $\angle BOD$) and $\angle BCD = \frac{1}{2}\angle BOD$ (2). Sum $= \frac{1}{2} \times 360^\circ = 180^\circ$ (1).	2+1
	OR: Half-chord $= \sqrt{15^2 - 9^2} = 12$; chord $= 24$ cm.	2+1
18	Distance of AB: $\sqrt{17^2 - 15^2} = 8$; of CD: $\sqrt{17^2 - 8^2} = 15$ (2). Same side ⇒ $15 - 8 = 7$ cm (1).	2+1
	OR: Distances: $\sqrt{25 - 9} = 4$ and $\sqrt{25 - 16} = 3$ (2). Opposite sides: $4 + 3 = 7$ cm (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Join D to C and extend to E on arc AFB (1). $\triangle DCB$ is isosceles ($CB = CD$) ⇒ $\angle CBD = \angle CDB$; exterior angle $\angle BCE = 2\angle BDC$ (2). Likewise $\angle ACE = 2\angle CDA$ (1). Add: $\angle BCA = 2(\angle BDC + \angle CDA) = 2\angle BDA$ (1).	1+2+1+1
	OR: Let ABCD be cyclic with centre O. Arc BCD: A is outside it, so $\angle BAD = \frac{1}{2}$(reflex $\angle BOD$) (2). Arc BAD: C is outside it, so $\angle BCD = \frac{1}{2}\angle BOD$ (2). Sum $= \frac{1}{2} \times 360^\circ = 180^\circ$ (1).	2+2+1
20	Let $AB > DE$ with perpendiculars CF, CG from the centre C. $AC = CD$ (radii). By Pythagoras: $CF^2 + AF^2 = CG^2 + GD^2$ (2). F and G are midpoints, so $AF > GD$ (1) ⇒ $AF^2 > GD^2$ ⇒ $CF^2 < CG^2$ (1) ⇒ $CF < CG$ (1).	2+1+1+1
	OR: (i) 14 m path: $\sqrt{25^2 - 7^2} = 24$ m. 48 m path: $\sqrt{25^2 - 24^2} = 7$ m (2). (ii) The 48 m path (longer chord is nearer) (1). (iii) $24 + 7 = 31$ m (1). (iv) $24 - 7 = 17$ m (1).	2+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME — SET B

Chapter 5: **Circles** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 8 cm	half-chord $= \sqrt{5^2 - 3^2} = 4$, so chord $= 2 \times 4 = 8$
2	**(a)** 24 cm	half-chord $= \sqrt{15^2 - 9^2} = 12$, so chord $= 2 \times 12 = 24$
3	**(d)** 13 cm	$r = \sqrt{12^2 + 5^2} = 13$
4	**(b)** 25 cm	$r = \sqrt{7^2 + 24^2} = 25$
5	**(d)** 24 cm	$d = \sqrt{25^2 - 7^2} = 24$
6	**(d)** 42°	angle at the centre is double the angle on the circle
7	**(c)** 88°	opposite angles of a cyclic quadrilateral add to 180°
8	**(d)** 82°	central angle is double
9	**(a)** the diameter	it passes through the centre
10	**(d)** longer	longer chord $\Leftrightarrow$ closer to the centre

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	90° — the angle subtended by a diameter at any point of the circle (angle in a semicircle).	1+1
12	The perpendicular from the centre bisects the chord; by the Baudhāyana–Pythagoras Theorem half-chord $= \sqrt{r^2 - d^2}$, so chord $= 2\sqrt{r^2 - d^2}$.	1+1
13	$2x + 10 + 3x - 20 = 180 \Rightarrow 5x = 190 \Rightarrow x = 38$; $\angle P = 86^\circ, \angle R = 94^\circ$.	1+1
14	$\angle C = 180^\circ - 75^\circ = 105^\circ$; $\angle D = 180^\circ - 110^\circ = 70^\circ$.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Let $AB = CD$ with perpendiculars OM, ON from O. Then $AM = \frac{1}{2}AB = \frac{1}{2}CD = CN$ (1). In right triangles OMA, ONC: $OA = OC$, $AM = CN$ ⇒ congruent (RHS) (1), so $OM = ON$ (1).	1+1+1
16	B lies on the circle outside arc ADC, so the angle at the centre is double: $2 \times 110^\circ = 220^\circ$.	2+1
17	$\angle BOC = 2 \times 50^\circ = 100^\circ$ (1). $OB = OC \Rightarrow \angle OBC = \angle OCB = \frac{180^\circ - 100^\circ}{2} = 40^\circ$ (2).	1+2
	OR: $\angle ACB = 90^\circ$ (angle in a semicircle) (1); $\angle ABC = 180^\circ - 90^\circ - 35^\circ = 55^\circ$ (2).	1+2
18	Let chords AB, CD (equal) meet at P and draw $OM \perp AB, ON \perp CD$; $OM = ON$ (equal chords) (1). Right triangles OMP, ONP are congruent (RHS) ⇒ $PM = PN$ (1). Since $AM = CN$ (halves of equal chords), $AP = CP$ and so $PB = PD$ (1).	1+1+1
	OR: The diagonal is $\sqrt{5^2 + 12^2} = 13$; the quadrilateral is made of two right triangles with legs 5 and 12 (1+1): area $= 2 \times \frac{1}{2} \times 5 \times 12 = 60$ square units (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $BC = \sqrt{64 + 36} = 10$ (1). (ii) $\angle A = 90^\circ$ ⇒ BC is a diameter; centre $(4, 3)$, radius 5 (2). (iii) Distance from $(4, 3)$ to $(8, 6) = \sqrt{16 + 9} = 5$ (1). (iv) A rectangle; opposite angles are $90^\circ + 90^\circ = 180^\circ$ so it is cyclic (1).	1+2+1+1
	OR: Draw the circle through A, B, D (1). Suppose it cuts CD at E. ABED is cyclic ⇒ $\angle BAD + \angle BED = 180^\circ$ (1). Given $\angle BAD + \angle BCD = 180^\circ$ ⇒ $\angle BED = \angle BCD$ (1). But $\angle BED$ is the exterior angle of $\triangle BEC$ so $\angle BED > \angle BCD$ — a contradiction (2). Hence C lies on the circle.	1+1+1+2
20	Let AB be a chord with midpoint M and centre O. $OA = OB$ (radii), $AM = MB$, OM common ⇒ $\triangle OMA \cong \triangle OMB$ (SSS) (2). So $\angle OMA = \angle OMB = 90^\circ$ (1) ⇒ $OM \perp AB$ at its midpoint, i.e. O lies on the perpendicular bisector of AB (2).	2+1+2
	OR: (i) $A + C = 180$: $8x - 20 = 180 \Rightarrow x = 25$; $B + D = 180$: $6y = 180 \Rightarrow y = 30$ (2). (ii) $\angle A = 80^\circ, \angle B = 70^\circ, \angle C = 100^\circ, \angle D = 110^\circ$ (2). (iii) $80 + 70 + 100 + 110 = 360$ ✓ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option

attempted. Deduct up to ½ mark for missing units or figures where required.

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